

CS 556 Homework - Probability Theory

Solve the following questions. Type your solutions using L^AT_EX or any other program you prefer and submit your typed solutions as a PDF file.

Question 1 (20 points)

Top 7 cards are drawn from a well-shuffled standard 52-card deck. Find probability that:

- a) The 7 cards include exactly 3 aces.
- b) The 7 cards include exactly 2 kings.
- c) The 7 cards include exactly 3 aces. or exactly 2 kings, or both.

Question 2 (20 points)

Alice and Bob have $2n + 1$ coins, each coin with probability of heads equal to $1/2$. Bob tosses $n + 1$ coins, while Alice tosses the remaining n coins. Assuming independent coin tosses, show that the probability that after all coins have been tossed, Bob will have gotten more heads than Alice is $1/2$.

Question 3 (20 points)

We are given three coins: one has heads in both faces, the second has tails in both faces, and the third has a head in one face and a tail in the other. We choose a coin at random, toss it, and the result is heads. What is the probability that the opposite face is tails?

Question 4 (20 points)

Each of k jars contain m white and n black balls. A ball is randomly chosen from jar 1 and transferred to jar 2, then a ball is randomly chosen from jar 2 and transferred to jar 3, etc. Finally a ball is randomly chosen from jar k . Show that the probability that the last ball is white is the same as probability that the first ball is white, i.e. it is $m/(m+n)$.

Question 5 (20 points)

A power utility can supply electricity to a city from n different power plants. Power plant i fails with probability p_i ($p_i \forall i$ need not be the same), independent of the others.

- a) Suppose that any one plant can produce enough electricity to supply the entire city. What is the probability that the city will experience a black-out?
- b) Suppose that two power plants are necessary to keep the city from a blackout. Find the probability that the city will experience a blackout.